

ECE 573
Cloud Computing and Cloud Native Systems
Lecture 01 Introduction

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Outline

Administrative Issues

Introduction to Cloud Computing

Reading Assignment

- ▶ This lecture: Course Syllabus
- ▶ Next lecture: 1,6

Outline

Administrative Issues

Introduction to Cloud Computing

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- ▶ Course website:
<https://wngjia.github.io/ece573-web/>

Textbooks

- ▶ Required Textbook
 - ▶ “Cloud Native Go: Building Reliable Services in Unreliable Environments”, Matthew Titmus, O'Reilly Media, 2021.
- ▶ Plus additional research papers

Useful Websites

- ▶ `https://www.youtube.com`
 - ▶ Find tutorials.
- ▶ `http://stackoverflow.com`
 - ▶ Learn how to communicate with professionals.

Prerequisite

- ▶ Computer programming
 - ▶ Branch and loop, function
 - ▶ Class
 - ▶ Searching and sorting
 - ▶ Debugging and troubleshooting
- ▶ Computer organization
 - ▶ Memory and pointer
 - ▶ Interrupts
 - ▶ I/O subsystems for storage and communication

Software system design for cloud

- ▶ Cloud computing architecture
 - ▶ Virtualization, containerization, and resource management
 - ▶ Distributed storage, communication, and computing systems
- ▶ Cloud native system design and implementation
 - ▶ Scalability, resilience, manageability, observability
 - ▶ Go language introduction and common cloud native patterns

Homeworks/Projects

- ▶ 5 Homeworks
 - ▶ 2 points each for a total of 10 points
- ▶ 6 Projects
 - ▶ 20 points each for a total of 120 points
- ▶ Submit online in Canvas only.

ECE 573 Grading

- ▶ A: 115
- ▶ B: 100
- ▶ C: 80

Expectations and Late Policy

- ▶ This is a project based course and you are expected to practice your problem solving and troubleshooting skills.
- ▶ Learn how to manage deadlines. **Do NOT expect to finish a project during the weekend right before its deadline.**
- ▶ Late homeworks and projects will NOT be graded, unless
 - ▶ A request to extend the deadline is received by email **48 hours BEFORE the deadline.**
 - ▶ With 48 hours of the deadline or after, the request should be accompanied by **extraordinary reasons with documented proof like doctor's notes, or it will be rejected.**
- ▶ **Extraordinary reasons do NOT include**
 - ▶ Lost of code or progress or report due to software, hardware, and networking failures.
 - ▶ AI assistants generate code that doesn't work.
- ▶ Resubmission/regrading of projects are not allowed.

Project Setup

- ▶ A recent Windows computer with 4 CPU cores, 16GB memory, and 512GB SSD.
 - ▶ Or access to a x64 Ubuntu server with 4 CPU cores, 8GB memory, and 100GB storage.
 - ▶ We are not able to support ARM-based computers, like Apple MacBooks and Raspberry Pi's, since many docker images don't support ARM.
 - ▶ We are not able to support computers that are more than 5 years old since virtualization would require substantial amount of computational and I/O power.
- ▶ Internet access is required to access common code and package repositories like GitHub and Docker Hub.

How to survive succeed in this course?

- ▶ Read: all instructions are in written.
 - ▶ Tutorials, source code, documents, and **don't overlook command outputs**.
- ▶ Communicate: we are very happy to solve any issue you may meet but you need to let us know what's wrong.
 - ▶ <https://stackoverflow.com/help/how-to-ask>
- ▶ Learn to use AI assistants.
 - ▶ <https://www.deeplearning.ai/short-courses/chatgpt-prompt-engineering-for-developers/>
- ▶ Feel free to explore new computer hardware and software but make sure they do not interfere with your schedule to meet deadlines.

Ethics (Very Seriously)

- ▶ Read “IIT Code of Academic Honesty” and “IEEE Code of Conduct” (posted on the course website).
 - ▶ Projects/homeworks should be done individually.
 - ▶ Discussions on homeworks/projects are encouraged.
 - ▶ Interactions with AI assistants (prompts and answers) should not be shared since they are considered as your own work.
 - ▶ Source code from the lectures and instructions in this course can be used directly.
 - ▶ Source code from other online sources not directly related to this course may be used with proper references.
- ▶ All other writings and code should be **BY YOURSELF**.
 - ▶ NEVER SHARE YOUR WRITINGS/CODE WITH OTHERS!
 - ▶ NEVER USE WRITINGS/CODE FROM OTHERS!
 - ▶ NEVER POST YOUR PROJECT CODE OR ASK FOR HELP DIRECTLY ONLINE!
- ▶ Please review our **Academic Honesty Guidelines**.

<https://www.iit.edu/academic-affairs/academic-honesty-guidelines>

Outline

Administrative Issues

Introduction to Cloud Computing

Meeting Personal Computing Needs

- ▶ Buy a desktop or laptop computer that meets the need.
 - ▶ Or, source parts and assemble a computer by yourself.
- ▶ Things you have taken for granted.
 - ▶ A place to use the computer.
 - ▶ Power that is readily available.
 - ▶ Internet connection that is also available.

Meeting Business Computing Needs

- ▶ Needs are more diverse than those for personal computing.
- ▶ Demand of computing power beyond a single computer.
 - ▶ Demand may vary, e.g. peak vs off-peak times.
- ▶ Facility to hold many computer servers physically.
 - ▶ Space and cooling.
 - ▶ Power delivery.
 - ▶ Network bandwidth.
- ▶ Redundancy in case of power and network failures.
- ▶ Have multiple facilities across the world.
 - ▶ Better performance to serve users locally.
 - ▶ Better redundancy in case of extreme events.
- ▶ Will existing software running on a single computer adapt automatically with multiple servers?

Rise of Cloud Computing

- ▶ Supercomputers for high-performance computing (HPC)
 - ▶ Primarily target at R&D of national interest that requires substantial computational power.
 - ▶ Specially made computers that are highly-reliable.
 - ▶ Need special skills to develop HPC software.
 - ▶ Very high up-front cost, difficult to expand.
- ▶ 2004: Google MapReduce
 - ▶ A programming model and framework to simplify software development for parallel data processing.
 - ▶ Run on cluster of commodity computer servers.
 - ▶ Automatically handle server failures.
 - ▶ Allow to expand cluster with more servers as business grows.
- ▶ 2006: Amazon Web Services
 - ▶ Including Amazon S3 for storage and EC2 for computing.
 - ▶ Allow businesses to rent computing resources on demand.
 - ▶ Without high up-front cost and long time to build physical infrastructure.

Why Cloud Computing?

- ▶ An economically viable solution for business computing needs.
- ▶ Economies of scale in cloud computing
 - ▶ Fixed costs for facilities are shared among customers.
 - ▶ Bulk purchasing power allows cloud providers to negotiate better deals with suppliers on servers etc.
 - ▶ Diverse range of customers smooth the demand of computing power over time and increase resource utilization.
 - ▶ Operation of large number of servers requires automated tools to reduce human errors, which in turn reduces labor costs.
- ▶ Open-source movement reduces cost of cloud software development
 - ▶ No software license fee
 - ▶ Encourage interoperability
 - ▶ Reduce cost to train new workforce

Categories of Cloud Services

- ▶ Infrastructure as a Service (IaaS)
 - ▶ Developers rent virtualized computing resources like virtual machines, storage, and networking components.
 - ▶ Allow developers to have complete control over software stack including networking, OS, library, and application.
- ▶ Platform as a Service (PaaS)
 - ▶ Developers rent pre-configured tools, libraries, and services like databases and message queues.
 - ▶ Allow developers to focus on core business logic while maintenance of common software components are outsourced.
- ▶ Software as a Service (SaaS)
 - ▶ End-users rent applications like email that can be accessed anywhere via a browser or a dedicated client.
 - ▶ Developers manage operation and maintenance of the software application to avoid compatibility issues with end-users' hardware and software systems.
 - ▶ End-users no longer need to worry about updates and upgrades other than their own devices.

Pay-per-Use vs. Subscription-based Pricing

- ▶ IaaS and PaaS are usually charged pay-per-use.
 - ▶ Cores/memory/storage are charged per quantity per hour.
 - ▶ Network traffics are charged per bytes depending on direction.
 - ▶ Database queries are charged per each depending on functions.
 - ▶ However, it may become very complicated to estimate the amount of usage and thus charge for actual applications.
- ▶ Virtual Private Servers (VPS) provide subscription-based IaaS
 - ▶ Pre-defined packages of cores, memory, storage, and networking resources for a fixed monthly or yearly fee.
 - ▶ A good option to try cloud computing for beginners.
- ▶ SaaS are usually subscription-based.
 - ▶ End-users have a clear understanding of the cost.
 - ▶ Providers have a predictable and steady revenue stream, making it easier to plan and allocate resources.
 - ▶ Perpetual licensing is not a good fit when software needs to be updated and upgraded often.
- ▶ Overall, pay-per-use pricing allows to optimize usage to reduce costs while subscription-based pricing is more predictable.

Summary

- ▶ Cloud computing provides an economically viable solution for business computing needs.